Pearson Edexcel International A Level · Pure Mathematics 2

Chapter 7 — Differentiation
Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P2 第7章 · 微分应用
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

1. P2-C07-R01 · P2 · Chapter 7

The curve C has equation $y = \frac{1}{4}x^4 - x^3$.

(a) Find the coordinates of all stationary points. (3)

(b) Using the sign of the first derivative,

(i) determine which stationary point is a minimum and explain why the other is not a turning point. (2)

(ii) state the intervals on which the curve is decreasing. (1)

(Total for Question 1 is 6 marks)

$f(x) = x^4 - 8x^2 + 3$.

(a) Find and classify all stationary points. (4)

(b) Find the intervals on which f is strictly increasing. (2)

(Total for Question 2 is 6 marks)

3.

A rectangle has perimeter 40 m. Its width is x m, with $0 < x \leq 10$.

(a) Find its area $A(x)$ and stationary value. (3)

(b) Use differentiation to justify that this is the maximum on the permitted domain. (3)

(Total for Question 3 is 6 marks)

4.

For $x > 0$, $C(x) = x^2 + 16/x$ is a cost model.

(a) Find the stationary point. (3)

(b) Show that this gives the global minimum for $x > 0$. (3)

(Total for Question 4 is 6 marks)

5. P2-C07-R05 · P2 · Chapter 7

$f(x) = 2x^3 - 9x^2 + 12x - 1$ on $[0, 3]$.

(a) Find its stationary points. (3)

(b) Find the greatest and least values on the stated closed interval. (3)

(Total for Question 5 is 6 marks)

6.

The curve $C : y = x^3 + ax^2 + bx$ has stationary points with $x = -1$ and $x = 3$.

(a) Find a, b. (3)

(b) Find and classify the two stationary points. (3)

(Total for Question 6 is 6 marks)

A box without a lid is made by removing squares of side x cm from each corner of a 12 cm by 18 cm sheet and folding up the sides.

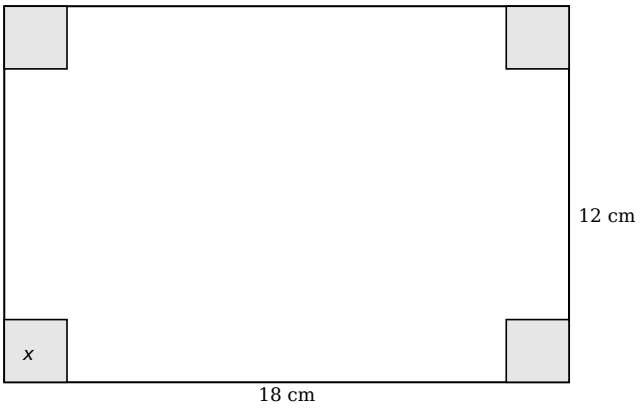


Figure 7

(a) Show that $V = 4x^3 - 60x^2 + 216x$ and state its physical domain. (4)

(b) Find the cut size that maximises volume, giving an exact answer and a maximum-volume justification. (4)

(Total for Question 7 is 8 marks)

8. P2-C07-R08 · P2 · Chapter 7

$f(x) = x^3 - 3x^2 + 3x + 4.$

(a) Find its stationary point. (2)

(b) Use the sign of f' on either side to show that this point is neither a local maximum nor a local minimum. (3)

(Total for Question 8 is 5 marks)

9. P2-C07-R09 · P2 · Chapter 7

For $x > 0$, $A(x) = 12x^{1/2} - x^{3/2}$.

(a) Find the stationary value of x. (3)

(b) Show that the corresponding value is a maximum and find it. (3)

(Total for Question 9 is 6 marks)

A product sells for £p per item. Weekly demand is modelled by $q = 300 - 20p$, for $0 < p < 15$.

(a) Find the revenue in terms of p and the price giving maximum revenue. (3)

(b) At this price, find

(i) the weekly revenue, (2)

(ii) the demand and a reason why the stationary revenue is a maximum. (3)

(Total for Question 10 is 8 marks)

$C : y = x^3 - 12x + 1.$

(a) Find the intervals where C has negative gradient. (2)

(b) Find the exact maximum and minimum values of y on $[-3, 3]$. (3)

(Total for Question 11 is 5 marks)

A cylinder has fixed volume 54π cm^3, radius $r > 0$ and height h. The total surface area of a closed cylinder is $2\pi r^2 + 2\pi r h$.

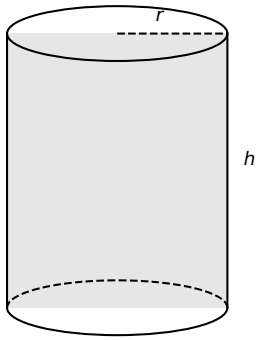


Figure 12

(a) Show that $S(r) = 2\pi r^2 + 108\pi / r$. (3)

(b) Find the radius and height giving minimum surface area and justify the minimum. (4)

(Total for Question 12 is 7 marks)

$f(x) = x^4 + ax^2 + b$. It is claimed to have a stationary point at $(2, -3)$ and to satisfy $f(0) = 5$.

(a) Determine whether these three conditions are consistent. (4)

(b) If $(2, -3)$ is replaced by $(2, -11)$, find and classify all stationary points. (4)

(Total for Question 13 is 8 marks)

A rectangular pen is built against a straight wall. Fencing of length 30 m is used for three sides only.

(a) Form an area function in the width x m perpendicular to the wall, and state its domain. (3)

(b) Find the maximum area and the corresponding dimensions. (4)

(Total for Question 14 is 7 marks)

15. P2-C07-R15 · P2 · Chapter 7

The curve C has equation $y = x^2 + 4/x$ for $x > 0$.

(a) Find its stationary point exactly. (4)

(b) Find the minimum gradient of C on $[1, 2]$. (3)

(Total for Question 15 is 7 marks)

16. P2-C07-R16 · P2 · Chapter 7

$f(x) = x^3 - 6x^2 + cx$. It has a local maximum at $x = 1$.

(a) Find c and verify the local maximum condition. (4)

(b) Find the other stationary point and the vertical difference between the stationary points. (3)

(Total for Question 16 is 7 marks)

17.

A rectangular sheet has area $48\ \text{cm}^2$. Its sides are x cm and $48/x$ cm, with $x > 0$.

(a) Find the perimeter $P(x)$ and its derivative. (3)

(b) Find the minimum perimeter in exact form and justify it. (4)

(Total for Question 17 is 7 marks)

18. P2-C07-R18 · P2 · Chapter 7

The curve is $y = x^5 - 5x^4$.

(a) Find the stationary points. (3)

(b) Use a sign table for y' to classify them. (4)

(Total for Question 18 is 7 marks)

19. P2-C07-R19 · P2 · Chapter 7

A storage box has square base of side x m and height h m. It has no lid and its total material area is $75\ \text{m}^2$.

(a) Express the volume as $V(x) = \frac{75}{4}x - \frac{1}{4}x^3$ and state the permitted interval. (4)

(b) Find the maximum possible volume and dimensions. (4)

(Total for Question 19 is 8 marks)

$f(x) = 2x^3 + 3x^2 - 12x + 7.$

(a) Find the intervals where f is decreasing. (3)

(b) Find the equation of the tangent at each stationary point. (3)

(Total for Question 20 is 6 marks)

A model for daily profit is $P = 60q - 3q^2 - 100$, where q is the number of items made and q is a non-negative integer.

(a) Use differentiation to find the optimal production level and maximum profit. (4)

(b) Find all integer values of q for which profit is at least £197. (3)

(Total for Question 21 is 7 marks)

The curve C is $y = x^3 + 3/x$ for $x > 0$.

(a) Find its stationary point exactly. (3)

(b) Prove that C has no local maximum for $x > 0$. (3)

(Total for Question 22 is 6 marks)

23. P2-C07-R23 · P2 · Chapter 7

A rectangular poster has printed area 96 cm^2. Margins of 2 cm are added on each vertical edge and 3 cm on each horizontal edge. The printed width is $x > 0$.

(a) Show that the total poster area is $A = 120 + 6x + 384/x$. (3)

(b) Find the printed width and height giving minimum total area. (3)

(Total for Question 23 is 6 marks)

$C : y = ax^3 + bx^2 + 4$, where $a \neq 0$. It has stationary points at $x = 0$ and $x = 2$, and passes through $(2, 0)$.

(a) Find a, b. (4)

(b) Classify each stationary point and find the greatest value on $[-1, 3]$. (3)

(Total for Question 24 is 7 marks)

A square-based closed box has volume $32\ \text{cm}^3$. Its base side is $x > 0$ cm.

(a) Find its total surface area in terms of x. (3)

(b) Find the dimensions of the box with least surface area and the least area. (4)

(Total for Question 25 is 7 marks)

$f(x) = x^3 - 3kx$, where $k > 0$.

(a) Find the stationary points in terms of k. (3)

(b) The difference between their y-coordinates is 32. Find k. (3)

(Total for Question 26 is 6 marks)

A curve is $y = x^4 - 4ax^2$, where $a > 0$.

(a) Find the three stationary x-coordinates. (3)

(b) Find a if the local minimum value is -36. (3)

(Total for Question 27 is 6 marks)

A rectangular enclosure has area $72\ \text{m}^2$ and one internal fence parallel to its width. Let its width be $x > 0$ m.

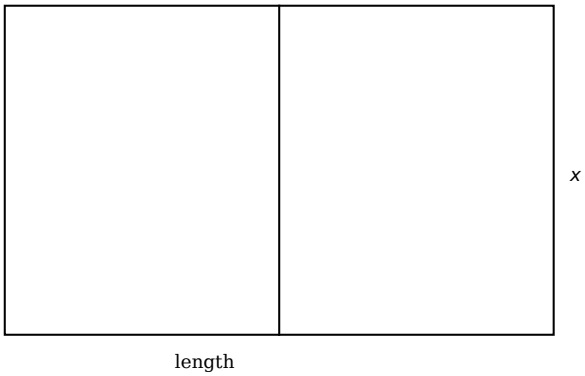


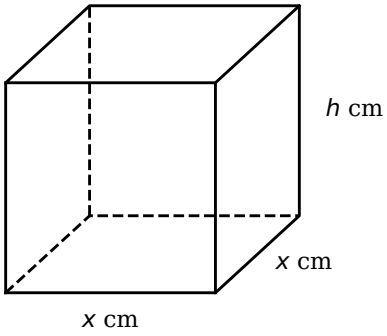
Figure 28

(a) Find the total fence length $L(x)$. (3)

(b) Find the dimensions giving minimum fence length and this minimum exactly. (4)

(Total for Question 28 is 7 marks)

A container has a square horizontal base of side x cm and height h cm. It has a base and four vertical sides but no lid. Its internal volume is 108 cm^3. The material used for its base costs twice as much per square centimetre as the material for its sides. Thickness and joins are ignored.



Open top; square base. Not to scale.

Figure 29

(a) Show that the total material cost is proportional to $C(x) = 2x^2 + 432/x$. (3)

(b) For $x > 0$,

(i) find the dimensions giving minimum cost. (3)

(ii) justify that the stationary value is a minimum. (2)

(Total for Question 29 is 8 marks)

The gradient of a curve is $dy/dx = 3x^2 - 12x + 9$. The curve passes through $A(0, 2)$.

(a) Find the equation of the curve. (2)

(b) For this curve, find

(i) the coordinates of both stationary points and determine their nature. (4)

(ii) the exact area between the curve and the line $y = 2$ for $0 \leq x \leq 3$. (2)

(Total for Question 30 is 8 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P2-C07-R01 · 6 marks

(a) · 3 marks

$(0,0)$ and $(3,-27/4)$.

$y\prime = x^3 - 3x^2 = x^2(x-3)$；零点$0,3$，代入曲线得两坐标。

M1	求导并分解
A1	两横坐标
A1	两纵坐标

(b) Using the sign of the first derivative,

(i) · 2 marks

$(3,-27/4)$ is a minimum; $(0,0)$ is stationary but not a turning point.

x^2除零点外为正，所以导数符号主要由$x-3$决定：在0两侧均负，无增减改变；在3左负右正，故为极小。

M1	正确区间符号
A1	两种分类与理由

(ii) · 1 marks

$(-\infty,0)$ and $(0,3)$.

要求导数严格小于零，需$x<3$且$x\neq 0$；分写成两个开区间。

B1	严格负导数区间

Question 2

P2-C07-R02 · 6 marks

(a) · 4 marks

$(0, 3)$ local maximum; $(\pm 2, -13)$ local minima.

f′=4x(x−2)(x+2)，f″=12x²−16，在0负，在±2正。

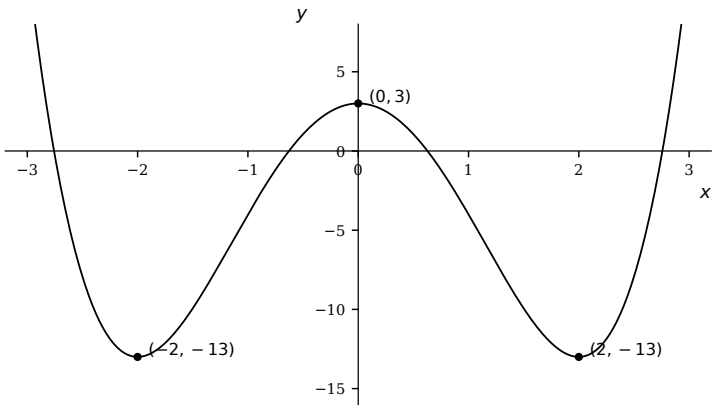
M1	一阶零点
M1	回代
M1	二阶符号
A1	三点分类

(b) · 2 marks

$(-2, 0)$ and $(2, \infty)$.

导数4x(x²−4)>0，在−2到0和2以上为正。

M1	导数符号分段
A1	两区间



Suggested sketch

Question 3

P2-C07-R03 · 6 marks

(a) · 3 marks

$A = 20x - x^2$; stationary at $x = 10$, area $100\ \text{m}^2$.

长20−x，导数20−2x=0。

M1	正确面积模型
M1	求导找点
A1	面积

(b) · 3 marks

$A'' = -2 < 0$ and $A' = 20 - 2x > 0$ on $0 < x < 10$, so maximum at 10.

二阶导负，且直到可行端点都递增，最大点是允许的正方形。

M1	二阶负或全区间正导数
M1	使用可行端点
A1	最大性

Question 4

P2-C07-R04 · 6 marks

(a) · 3 marks

$x = 2$, $C = 12$.

C′=2x−16/x²=0给x³=8，代C。

M1	正确负幂求导
M1	解正根
A1	点

(b) · 3 marks

$C' = 2(x^3 - 8)/x^2$ is negative for $0 < x < 2$ and positive for $x > 2$.

分母正，导数符号由x³−8决定，先降后升遍及整个正域。

M1	正确导数因式符号
M1	两侧符号
A1	全局最小

Question 5

P2-C07-R05 · 6 marks

(a) · 3 marks

$(1, 4)$ and $(2, 3)$.

f′=6(x−1)(x−2)，代原式。

M1	导数因式
M1	求零点
A1	两点

(b) · 3 marks

Greatest 8 at $x = 3$; least -1 at $x = 0$.

比较f0=−1、f1=4、f2=3、f3=8；必须包括端点。

M1	列出驻点及端点候选
M1	正确四值
A1	最大最小及位置

Question 6

P2-C07-R06 · 6 marks

(a) · 3 marks

$a = -3, b = -9$.

导函数首项$3x^2$，已知两根所以$y'=3(x+1)(x-3)=3x^2-6x-9$；比较2a、b。

M1	导数根构造
M1	比较系数
A1	a与b

(b) · 3 marks

$(-1, 5)$ local maximum; $(3, -27)$ local minimum.

$y''=6x-6$，-1时负、3时正；原函数代值。

M1	两纵坐标
M1	二阶符号
A1	两点类型

Question 7

P2-C07-R07 · 8 marks

(a) · 4 marks

$V = x(12-2x)(18-2x);\ 0 < x < 6.$

底边分别减2x，高x，三尺寸均须正。

M1	正确三尺寸
M1	展开
A1	多项式
B1	域

(b) · 4 marks

$x = 5 - \sqrt{7}$ cm; the other stationary root $5 + \sqrt{7} > 6$ is invalid, and $V'' = 24x - 120 < 0$ at the valid root.

V′=12x²−120x+216=12(x²−10x+18)，求两根，检查域及二阶导。

M1	正确求导方程
M1	两根与域筛选
M1	最大值检验
A1	精确x

Question 8

P2-C07-R08 · 5 marks

(a) · 2 marks

$(1, 5)$.

f′=3(x−1)²为0只在1，代曲线得5。

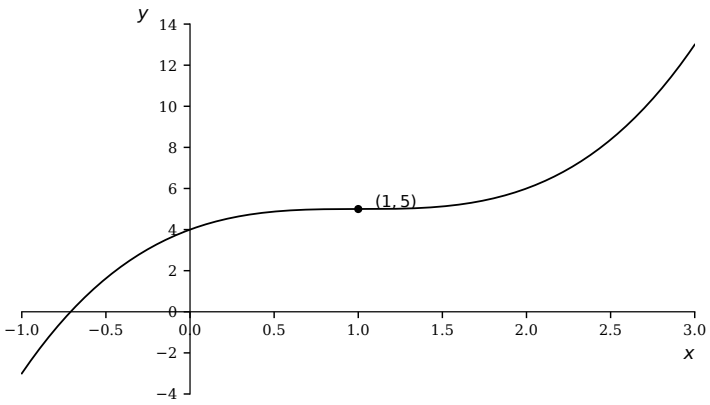
M1	求导并找根
A1	点

(b) · 3 marks

$f'(x) > 0$ for $x < 1$ and $x > 1$; the curve increases on both sides.

二阶导在1为0不能作极值判定；导数正号不变，函数没有转升为降或反之。

M1	两侧导数符号
M1	符号不变
A1	非极值结论



Suggested sketch

Question 9

P2-C07-R09 · 6 marks

(a) · 3 marks

$x = 4$.

A′=6x^{−1/2}−(3/2)x^{1/2}=0，乘2√x得12−3x=0。

M1	分数幂导数
M1	解正域方程
A1	x

(b) · 3 marks

Maximum 16; $A'' = -3x^{-3/2} - (3/4)x^{-1/2} < 0$.

二阶导在整个正域负；代x4得24−8=16。

M1	二阶导
M1	负性
A1	最大值

Question 10

P2-C07-R10 · 8 marks

(a) · 3 marks

$R = 300p - 20p^2$; $p = 7.5$.

收入单价乘销量，R′=300−40p=0。

M1	收入模型
M1	求导解驻点
A1	价格

(b) At this price, find

(i) · 2 marks

£1125.

q=150，收入7.5×150。

M1	模型代值
A1	收入

(ii) · 3 marks

150 items; $R'' = -40 < 0$.

代需求式；二阶导严格负，且价格在物理域。

B1	需求
M1	二阶导负
A1	最大性

Question 11

P2-C07-R11 · 5 marks

(a) · 2 marks

$-2 < x < 2$.

y′=3x²−12=3(x−2)(x+2)，向上二次式两根间负。

M1	求导与根
A1	区间

(b) · 3 marks

Maximum 17 at $x = -2$; minimum -15 at $x = 2$.

比较y(−3)=10,y(−2)=17,y2=−15,y3=−8。

M1	端点与驻点候选
M1	正确代值
A1	两极值与位置

Question 12

P2-C07-R12 · 7 marks

(a) · 3 marks

$h = 54/r^2$, hence the stated formula.

体积$\pi r^2 h$固定，先消h再代给定表面积式。

M1	体积条件消元
M1	代面积式
A1	简式

(b) · 4 marks

$r = 3$ cm, $h = 6$ cm; $S'' = 4\pi + 216\pi/r^3 > 0$.

$S' = 4\pi r - 108\pi/r^2 = 0$给$r^3 = 27$；二阶导正，再用体积求h。

M1	正确求导及正根
M1	最小性
A1	半径
A1	高

Question 13

P2-C07-R13 · 8 marks

(a) · 4 marks

$a=-8, b=5$ would give $f(2)=-11$, so these stated conditions are inconsistent.

f0给b5，f′2=32+4a=0给a−8；代f2=16−32+5=−11不等于−3。

M1	用原点值
M1	用驻点导数
M1	回检纵坐标
A1	一致性结论

(b) · 4 marks

$(0,5)$ maximum; $(\pm 2,-11)$ minima.

此时f=x^4−8x^2+5，f′=4x(x^2−4)，f″=12x^2−16，逐一判号。

M1	全部零点
M1	纵坐标
M1	二阶符号
A1	分类

Question 14

P2-C07-R14 · 7 marks

(a) · 3 marks

$A = x(30 - 2x),\ 0 < x < 15.$

两宽用2x，剩余长30−2x，几何尺寸正。

M1	三侧栅栏关系
M1	面积
A1	域

(b) · 4 marks

Area $112.5\ \text{m}^2$; width 7.5 m, length 15 m.

A′=30−4x=0给7.5，A″=−4证明最大；回求长和面积。

M1	求导解驻点
M1	最大性
A1	两尺寸
A1	面积

Question 15

P2-C07-R15 · 7 marks

(a) · 4 marks

$\left(\sqrt[3]{2}, 3\sqrt[3]{4}\right)$.

y′=2x−4/x²=0给x³=2；此时4/x=2x²，所以y=3x²。

M1	正确导数
M1	正根
M1	化简纵坐标
A1	点

(b) · 3 marks

-2 at $x = 1$.

梯度函数g=2x−4/x²，其导数g′=2+8/x³>0，所以g在区间递增，最小在1。

M1	区分梯度函数与原函数
M1	正二阶导说明梯度递增
A1	最小梯度及位置

Question 16

P2-C07-R16 · 7 marks

(a) · 4 marks

$c = 9$; $f''(1) = -6 < 0$.

f′1=3−12+c=0给9；再检二阶导6x−12。

M1	驻点条件
A1	c
M1	二阶负
A1	最大性

(b) · 3 marks

Other point $(3, 0)$; vertical difference 4.

导数3(x−1)(x−3)；f1=4，f3=0。

M1	另一零点
M1	原式代值
A1	点与纵差

Question 17

P2-C07-R17 · 7 marks

(a) · 3 marks

$P = 2x + 96/x$; $P' = 2 - 96/x^2$.

两边长之和乘2，再逐项求导。

M1	正确周长
M1	负幂求导
A1	导式

(b) · 4 marks

$16\sqrt{3}$ cm, at $x = 4\sqrt{3}$; $P'' = 192/x^3 > 0$.

令x^2=48取正根，另一边同长；二阶导正；周长四倍边长。

M1	解正驻点
M1	二阶最小性
M1	代周长
A1	精确值

Question 18

P2-C07-R18 · 7 marks

(a) · 3 marks

$(0,0)$ and $(4,-256)$.

$y'=5x^3(x-4)$，零点0、4；代原式。

M1	因式化导数
M1	两点代值
A1	坐标

(b) · 4 marks

$(0,0)$ local maximum; $(4,-256)$ local minimum.

$x<0$时两因子负故导数正；$0<x<4$正。0处二阶导零，须用符号变化。

M1	三段导数符号
M1	0处正转负
M1	4处负转正
A1	分类

Question 19

P2-C07-R19 · 8 marks

(a) · 4 marks

$x^2+4xh=75;\ V=x^2h=(75x-x^3)/4,\ 0<x<5\sqrt{3}.$

从材料面积消h，h正给$x^2<75$。

M1	正确无盖面积
M1	消元
A1	V式
B1	物理域

(b) · 4 marks

$x=5$ m, $h=5/2$ m; maximum $125/2$ m^3.

$V'=(75-3x^2)/4=0$给$x5$，$V''=-3x/2<0$，再求h和V。

M1	正驻点
M1	最大性
A1	尺寸
A1	体积

Question 20

P2-C07-R20 · 6 marks

(a) · 3 marks

$-2 < x < 1$.

f′=6(x+2)(x−1)，在两根间负。

M1	求导并分解
M1	符号
A1	区间

(b) · 3 marks

$y = 27$ at $x = -2$; $y = 0$ at $x = 1$.

f−2=−16+12+24+7=27；f1=2+3−12+7=0，驻点切线水平。

M1	两纵坐标
M1	水平切线性质
A1	两方程

Question 21

P2-C07-R21 · 7 marks

(a) · 4 marks

$q = 10$; maximum £200.

$P'=60-6q=0$，二阶$-6<0$，驻点10为允许整数，代值200。

M1	正确求导与驻点
M1	最大性及整数合法
A1	产量
A1	利润

(b) · 3 marks

$q = 9, 10, 11$.

$P=200-3(q-10)^2 \geqslant 197$给平方$\leqslant 1$，取三个非负整数。

M1	二次利润门槛
M1	解范围并取整数
A1	全部产量

Question 22

P2-C07-R22 · 6 marks

(a) · 3 marks

$(1, 4)$.

y′=3x²−3/x²=0给x⁴=1，正域取1。

M1	负幂导式
M1	解域内根
A1	点

(b) · 3 marks

$y'' = 6x + 6/x^3 > 0$ for $x > 0$, so its only stationary point is a minimum; the derivative changes from negative to positive.

二阶导正使梯度严格递增，唯一驻点为最小，不存在其他驻点最大。

M1	二阶导正
M1	唯一驻点或导数符号
A1	无最大结论

Question 23

P2-C07-R23 · 6 marks

(a) · 3 marks

$(x+4)(96/x+6)=120+6x+384/x$.

总宽加4，总高加6，展开并合并96+24。

M1	正确外尺寸
M1	展开
A1	A式

(b) · 3 marks

Width 8 cm, height 12 cm; $A''=768/x^3>0$.

A′=6−384/x²=0给x8，再高96/8=12；二阶导正。

M1	求正驻点
M1	最小性
A1	两尺寸

Question 24

P2-C07-R24 · 7 marks

(a) · 4 marks

$a = 1, b = -3$.

导数x(3ax+2b)，在2零给6a+2b=0；点条件8a+4b+4=0。

M1	驻点条件
M1	点条件
M1	联立
A1	系数

(b) · 3 marks

$(0, 4)$ maximum, $(2, 0)$ minimum; greatest 4 at $x = 0$ and $x = 3$.

二阶6x−6给两类型；端点f−1=0、f3=4，所以最大4也在x=3取得。

M1	二阶分类
M1	检查端点
A1	最大值及所有位置

Question 25

P2-C07-R25 · 7 marks

(a) · 3 marks

$S = 2x^2 + 128/x$.

h=32/x²，两底2x²加四侧4xh。

M1	体积消h
M1	两底四侧
A1	S式

(b) · 4 marks

$x = \sqrt[3]{32}$ cm, $h = x$; least area $6\,32^{2/3}$ cm^2.

S′=4x−128/x²=0给x³32；S″=4+256/x³正，h=x。

M1	正驻点
M1	二阶最小性
A1	尺寸
A1	精确面积

Question 26

P2-C07-R26 · 6 marks

(a) · 3 marks

$(-\sqrt{k}, 2k^{3/2})$ and $(\sqrt{k}, -2k^{3/2})$.

f′=3x²−3k=0，x=±√k，利用x²=k化纵坐标−2kx。

M1	正确驻点根
M1	化简纵坐标
A1	两点

(b) · 3 marks

$k = 4$.

纵差4k^{3/2}=32给k^{3/2}=8，k=8^{2/3}=4。

M1	正确纵差方程
M1	指数求解
A1	k

Question 27

P2-C07-R27 · 6 marks

(a) · 3 marks

$x = 0, \pm\sqrt{2a}.$

y′=4x(x²−2a)，三不同实根由a正保证。

M1	正确导数分解
M1	正根条件
A1	三x

(b) · 3 marks

$a = 3.$

在x²=2a时y=4a²−8a²=−4a²；−4a²=−36，a正取3。

M1	求极小纵坐标
M1	使用参数正号
A1	a

Question 28

P2-C07-R28 · 7 marks

(a) · 3 marks

$L = 3x + 144/x.$

外框两宽两长加一条内部宽，长度72/x。

M1	三宽两长计数
M1	面积消元
A1	L式

(b) · 4 marks

Width $4\sqrt{3}$ m, length $6\sqrt{3}$ m; minimum $24\sqrt{3}$ m.

L′=3−144/x²=0给x²48，正根4√3；L″=288/x³正，代回。

M1	求正驻点
M1	最小性
A1	两尺寸
A1	最小长度

Question 29

P2-C07-R29 · 8 marks

(a) · 3 marks

$h = 108/x^2$; $C = 2x^2 + 432/x$.

以侧面每平方厘米单价为单位，底面成本$2x^2$，四侧面成本$4xh$；由体积代入h。

M1	用体积消去h
M1	正确加权面积
A1	目标式

(b) For $x > 0$,

(i) · 3 marks

$x = \sqrt[3]{108}$ cm, $h = \sqrt[3]{108}$ cm.

$C\prime = 4x - 432/x^2 = 0$给$x^3 = 108$；由$h = 108/x^2$可知$h = x$。

M1	求导并设为零
A1	最优x
A1	对应h

(ii) · 2 marks

$C\prime\prime(x) = 4 + 864/x^3 > 0$ for $x > 0$.

二阶导数在所有正x处严格为正，特别在临界点处为正，所以得到最小成本。

M1	二阶导数
A1	正确最小值结论

Question 30

P2-C07-R30 · 8 marks

(a) · 2 marks

$y = x^3 - 6x^2 + 9x + 2$.

逐项积分并用$y(0) = 2$确定常数。

M1	正确积分
A1	常数

(b) For this curve, find

(i) · 4 marks

$(1, 6)$ maximum; $(3, 2)$ minimum.

导数$3(x-1)(x-3)$，零点$1, 3$；$y'' = 6x - 12$在两处分别为负、正，代入求纵坐标。

M1	找出临界横坐标
A1	两点坐标
M1	用二阶导数分类
A1	两种性质

(ii) · 2 marks

$27/4$.

$y - 2 = x(x-3)^2 \geq 0$，所以面积$\int_0^3 (x^3 - 6x^2 + 9x)\,dx = [x^4/4 - 2x^3 + 9x^2/2]_0^3 = 27/4$。

M1	正确上减下积分
A1	精确面积